Chapter 5: Sampling Distributions

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Outline

1. Introduction to Sampling Distributions
2. Distribution of Sample Mean
3. Distribution of Sample Variance
4. Distribution of Sample Proportion
5. Summary and Key Takeaways

What is a Sampling Distribution?

Definition (Sampling Distribution)

A **sampling distribution** is the probability distribution of a statistic (such as mean, variance, proportion) obtained from many random samples of the same size from a population.

Key Concept

- A **statistic** is a summary measure from a sample (e.g., $\overline{X}$, S^2, $\hat{p}$)
- A **parameter** is a summary measure from the population (e.g., μ, σ^2, p)
- The sampling distribution tells us how the statistic varies across different samples

Why Study Sampling Distributions?

Practical Importance:

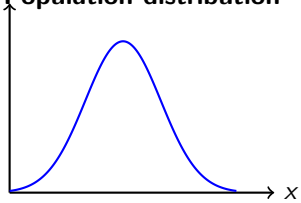
- Foundation for statistical inference
- Enables us to:
  - Construct confidence intervals
  - Perform hypothesis tests
  - Make predictions
- Quantifies uncertainty in estimates

Real-World Example:

- Population: All adults in a city
- Parameter: Mean height μ
- We take samples of size $n = 100$
- Each sample gives $\overline{X}$
- Sampling dist = distribution of all possible $\overline{X}$ values

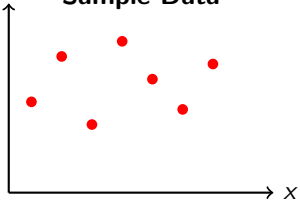
Three Important Distributions

Population distribution



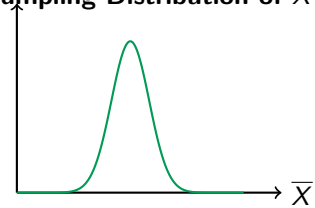
$f(x)$, parameters μ, σ^2

Sample Data



n observations, statistic $\overline{X}$

Sampling Distribution of $\overline{X}$



$\mathbb{E}(\overline{X}) = \mu$, $\mathrm{Var}(\overline{X}) = \sigma^2/n$

Sample Mean: Notation and Definition

Setup

- Population with mean μ and variance σ^2
- Random sample: $X_1, X_2, \ldots, X_n$ (i.i.d.)
- Sample mean: $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$

Theorem (Expected Value and Variance)

For any population with finite mean μ and variance σ^2:

$$\mathbb{E}(\overline{X}) = \mu \quad \textit{and} \ \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

$$SE(\overline{X}) = \frac{\sigma}{\sqrt{n}} \quad \textit{(Standard Error)}$$

Case 1: Normal Population

Theorem (Exact Distribution)

If $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ *(i.i.d.), then:*

$$\overline{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

for any sample size n.

Standardization

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

Why this matters:

- Can calculate probabilities exactly
- Foundation for t-tests and confidence intervals
- No minimum sample size requirement

Case 2: Non-Normal Population - Central Limit Theorem

Theorem (Central Limit Theorem (CLT))

Let $X_1, \ldots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2. Then, as $n \to \infty$:

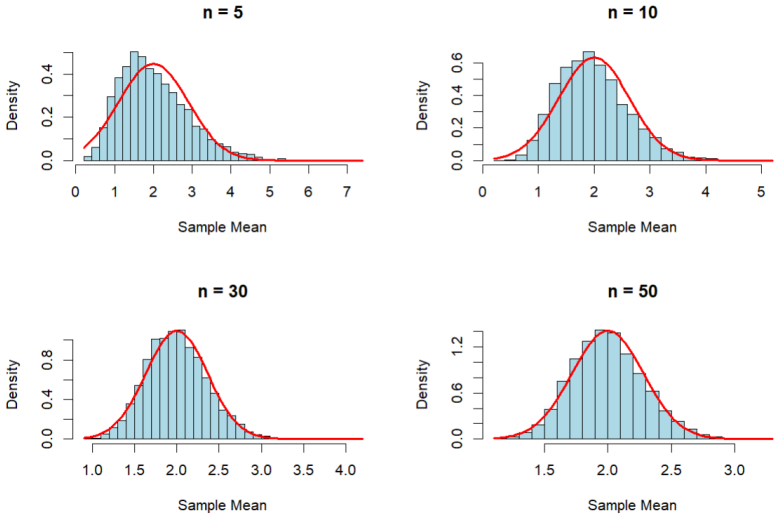
$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1)$$

Or equivalently: $\overline{X} \approx \mathcal{N}(\mu, \sigma^2/n)$ for large n.

Rule of Thumb

- $n \geq 30$: CLT usually works well
- Symmetric population: smaller n may suffice
- Highly skewed: need larger n (maybe $n \geq 50$)

CLT with different sample sizes



CLT Visualization: Effect of Sample Size

Source: Exponential Distribution (highly skewed)

Small n (n=5, n=10):

- Still somewhat skewed
- Wider distribution
- CLT approximation fair

Large n (n=30, n=50):

- Nearly perfect normal shape
- Much narrower
- CLT works excellently

Conclusion

Regardless of population shape, $\overline{X}$ becomes approximately normal as n increases!

R Tutorial: Calculating Probabilities

Problem: Heights of adults have $\mu = 170$ cm, $\sigma = 10$ cm. If we sample $n = 25$ people, what is $P(\overline{X} > 172)$?

```r
# Given information
mu <- 170
sigma <- 10
n <- 25
x_bar <- 172

SE <- sigma / sqrt(n) # Standard error

# Method 1: Using standardization
z <- (x_bar - mu) / SE
prob <- 1 - pnorm(z)

# Method 2: Direct calculation
prob_direct <- 1 - pnorm(x_bar, mean=mu, sd=SE)

# Verification by simulation
set.seed(123)
simulated_means <- replicate(10000, mean(rnorm(n, mean=mu, sd=sigma)))
empirical_prob <- mean(simulated_means > x_bar)
```

Sample Variance: Definition

Sample Variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

- Uses $n-1$ (degrees of freedom) instead of n
- Makes S^2 an unbiased estimator of σ^2: $\mathbb{E}(S^2) = \sigma^2$

Sample Standard Deviation

$$S = \sqrt{S^2}$$

Note: S is *not* unbiased for σ, but consistent.

Distribution of Sample Variance

Theorem (Chi-Square Distribution)

If $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ *(i.i.d.), then*

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

where χ^2_{n-1} *is the chi-square distribution with* $n-1$ *degrees of freedom.*

Properties of χ^2 Distribution

If $X \sim \chi^2_n$, then

- $\mathbb{E}(X) = n$ (degrees of freedom)
- $\mathrm{Var}(X) = 2n$
- Skewed to the right, approaches normal as n increases

Sample Proportion: Setup

Bernoulli Population

- Population of "success" (1) and "failure" (0)
- Proportion of successes: p
- Each observation: $X_i \sim \text{Bernoulli}(p)$

$$X_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

Sample Proportion

$$\hat{p} = \frac{\text{Number of successes}}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{X}{n}$$

where $X = \sum X_i \sim \text{Binomial}(n, p)$

Exact and Approximate Distributions

Theorem (Normal Approximation (CLT))

For large n, by CLT:

$$\hat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$

Or in standardized form:

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx \mathcal{N}(0, 1)$$

Condition for Normal Approximation

Works well when: $np \geq 10$ **and** $n(1-p) \geq 10$

Properties of Sample Proportion

Theorem (Mean and Variance)

$$\mathbb{E}(\hat{p}) = p$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

- $\hat{p}$ is unbiased for p
- Standard error is largest when $p = 0.5$
- As n increases, $\hat{p}$ becomes more precise
- $\text{Var}(\hat{p}) \to 0$ as $n \to \infty$

When Normal Approximation Fails

Effect of Sample Size and p

Good Approximation:

- $n = 100$, $p = 0.3$
- $np = 30 \geq 10$
- $n(1-p) = 70 \geq 10$
- Shape: Nearly normal

Poor Approximation:

- $n = 20$, $p = 0.1$
- $np = 2 < 10$
- $n(1-p) = 18 \geq 10$
- Shape: Skewed right

Warning

When p is close to 0 or 1, need larger n for normal approximation to work!

Applications of Sample Proportion

Real-World Examples

1. **Political Polling:**
   - Estimate proportion of voters supporting a candidate
   - Margin of error: $\approx 1.96 \times \sqrt{p(1-p)/n}$
2. **Quality Control:**
   - Proportion of defective items in a batch
   - Acceptance sampling plans
3. **Medical Research:**
   - Success rate of a treatment
   - Comparing two treatment proportions
4. **Market Research:**
   - Customer satisfaction rate
   - Click-through rate in advertising

Summary Table: Sampling Distributions

Statistic	**Distribution**	**Conditions**	**SE**
Sample Mean $\overline{X}$	$\mathcal{N}(\mu, \sigma^2/n)$ (approx. by CLT)	Pop. normal or $n \geq 30$	$\sigma/\sqrt{n}$
Sample Variance S^2	$(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$	Pop. normal	$\sigma\sqrt{2/(n-1)}$ (approx.)
Sample Proportion $\hat{p}$	$\mathcal{N}(p, p(1-p)/n)$ (approx. by CLT)	$np \geq 10$ and $n(1-p) \geq 10$	$\sqrt{p(1-p)/n}$

Key Takeaways

1. Central Concept

- Sampling distributions describe variability of statistics across samples
- Foundation for confidence intervals and hypothesis tests

2. Central Limit Theorem

- Works for any population (with finite variance)
- Sample size $n \geq 30$ usually sufficient
- Explains why normal distribution is so common

3. Standard Error

- Measures precision of estimator
- Decreases as n increases: $\text{SE} \propto 1/\sqrt{n}$
- To halve SE, need to quadruple sample size

R Functions Summary

Function	Purpose
`rnorm(n, mean, sd)`	Generate normal random variables
`mean(x)`	Calculate sample mean
`var(x)`	Calculate sample variance (n-1 formula)
`sd(x)`	Calculate sample standard deviation
`pnorm(q, mean, sd)`	Normal CDF $P(X \leq q)$
`qnorm(p, mean, sd)`	Normal quantile (inverse CDF)
`dnorm(x, mean, sd)`	Normal density function
`pchisq(q, df)`	Chi-square CDF
`qchisq(p, df)`	Chi-square quantile
`dchisq(x, df)`	Chi-square density
`rbinom(n, size, prob)`	Generate binomial random variables
`pbinom(q, size, prob)`	Binomial CDF
`replicate(n, expr)`	Repeat expression n times
`set.seed()`	Set random seed for reproducibility